LAB ASSIGNMENT 1

PROBLEM 1: Write a Program to take a 1D array and convert it to numpy and pandas

CODE:

```
import numpy as np
import pandas as pd
n = int(input())
arr = []
for i in range(n):
    arr.append(int(input()))
np_arr = np.array(arr)
pd_arr = pd.Series(arr)
print(np_arr)
print(pd_arr)
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB 1\LAB_1_a.py"
4
1
2
3
4
[1 2 3 4]
0    1
1    2
2    3
3    4
dtype: int64
```

PROBLEM 2: Write a Program to take a 2D array and convert it to a 1D array using numpy and pandas

CODE:

```
import numpy as np
import pandas as pd
r = int(input())
c = int(input())
arr = []
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    arr.append(row)
np_arr = np.array(arr).flatten()
pd_arr = pd.DataFrame(arr).values.flatten()
print(np_arr)
print(pd_arr)
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB 1\LAB_1_b.py"
2
2
1
2
3
4
[1 2 3 4]
[1 2 3 4]
```

PROBLEM 3: Write a Program to take 5 different coordinates in a loop and find the total error using linear regression

CODE:

```
import matplotlib.pyplot as plt
m = float(input("Enter value of m: "))
c = float(input("Enter value of c: "))
x_vals = []
y_vals = []
for i in range(5):
    x = float(input(f"Enter x coordinate {i+1}: "))
    y = float(input(f"Enter y coordinate {i+1}: "))
    x_vals.append(x)
    y_vals.append(y)
errors = []
y_calcs = []
total_error = 0
for i in range(5):
    y_calc = m * x_vals[i] + c
    y_calcs.append(y_calc)
    error =abs(y_vals[i] - y_calc)
    errors.append(error)
    total_error = total_error + error
print("\nErrors for each point:", errors)
print("Total Error:", total_error)
for i in range(5):
    plt.plot([x_vals[i], x_vals[i]], [y_vals[i], y_calcs[i]], 'k--')
plt.scatter(x_vals, y_vals)
plt.plot(x_vals, y_calcs)
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.xlabel("X")
plt.ylabel("Y")
plt.legend()
plt.show()
```

OUTPUT:

```
PS D:\NITJ ACADEMICS\OPENLEARN> python -u "d:\NITJ ACADEMICS\OPENLEARN\LAB_2026\LAB 1\LAB_1_c.py"
Enter value of m: 1
Enter value of c: 2
Enter x coordinate 1: 2
Enter y coordinate 1: 3
Enter x coordinate 2: 5
Enter y coordinate 2: 3
Enter x coordinate 3: 5
Enter y coordinate 3: 2
Enter x coordinate 4: 6
Enter y coordinate 4: 1
Enter x coordinate 5: 3
Enter y coordinate 5: 4

Errors for each point: [1.0, 4.0, 5.0, 7.0, 1.0]
Total Error: 18.0
```